

INTRODUCTION AND OPERATING SYSTEM STRUCTURES

Topics: Definition, Types of OS, Real-Time OS, System Components, System Services, System Calls, System Programs, System Structure, Virtual Machines, System Design & Implementation, System Generation

Subject Incharge: Mangnale M N

Subject : Operating System

1. Explain the concept of System Services provided by an Operating System.
2. Explain different types of Operating Systems (Batch, Multiprogramming, Time-Sharing, Real-Time, Distributed, Multiprocessor) with examples.
3. Explain Real-Time Operating Systems in detail. Differentiate between Hard Real-Time and Soft Real-Time systems.
4. What are System Calls? Explain the different categories of System Calls with suitable examples.
5. Explain the layered structure (or simple structure / modular structure) of an Operating System.
6. Explain the Virtual Machine concept and its advantages.
7. Discuss System Design and Implementation goals of an Operating System.
8. Explain System Generation (SYSGEN) in Operating Systems.
9. Define Operating System. Explain its functions and list any four types of OS with examples.

Multiple Choice Questions

1. **An Operating System is:**
 - A) A collection of programs that manages hardware resources
 - B) A hardware component
 - C) An application software
 - D) A programming language
2. **Which of the following is not a type of Operating System?**
 - A) Batch OS
 - B) Time-Sharing OS
 - C) Real-Time OS
 - D) Compiler OS
3. **In a Hard Real-Time system:**
 - A) Missing a deadline is acceptable
 - B) Missing a deadline can cause system failure
 - C) There are no deadlines
 - D) Deadlines are optional
4. **Which of the following is a system service provided by OS?**
 - A) Program execution
 - B) I/O operations
 - C) File system manipulation
 - D) All of the above
5. **System calls are used to:**
 - A) Communicate between user programs and OS kernel
 - B) Compile programs
 - C) Design hardware
 - D) None of the above
6. **Which of the following is not a category of system calls?**

- A) Process control
 - B) File management
 - C) Device management
 - D) Network design
7. **System programs provide:**
- A) A convenient environment for program development and execution
 - B) Direct hardware access
 - C) Only file management
 - D) None of the above
8. **In a layered OS structure, the bottommost layer is usually:**
- A) User interface
 - B) Hardware
 - C) Application programs
 - D) System calls
9. **A Virtual Machine provides:**
- A) An interface identical to the underlying bare hardware
 - B) Only software isolation
 - C) Faster hardware
 - D) None of the above
10. **The main advantage of Virtual Machines is:**
- A) Complete protection of system resources
 - B) Increased hardware cost
 - C) Slower execution
 - D) Limited portability
11. **In OS design, Mechanism determines:**
- A) How to do something
 - B) What will be done
 - C) When to do something
 - D) Who will do something
12. **Most modern operating systems are written in:**
- A) Assembly language only
 - B) High-level languages like C/C++
 - C) Machine language only
 - D) Java only
13. **System Generation (SYSGEN) is the process of:**
- A) Configuring OS for a specific hardware
 - B) Writing application programs
 - C) Compiling user programs
 - D) Debugging the kernel
14. **In a Time-Sharing Operating System:**
- A) Multiple users can share the computer simultaneously
 - B) Only one user can use the system
 - C) Jobs are processed in batches

- D) There is no multiprogramming
15. **Multiprogramming increases CPU utilization by:**
- A) Keeping multiple programs in memory
 - B) Running only one program at a time
 - C) Reducing memory size
 - D) Disabling interrupts
16. **The Kernel is:**
- A) The core part of the Operating System
 - B) An application program
 - C) A hardware device
 - D) A system utility
17. **Dual-Mode Operation provides:**
- A) Protection to the Operating System
 - B) Faster execution
 - C) More memory
 - D) Better graphics
18. **Which mode is used when the Operating System is executing?**
- A) User mode
 - B) Kernel mode (Supervisor mode)
 - C) Both modes simultaneously
 - D) Application mode
19. **Which of the following is an example of a Real-Time Operating System?**
- A) Windows 11
 - B) VxWorks
 - C) Ubuntu
 - D) macOS
20. **Policy in OS design decides:**
- A) How to do something
 - B) What will be done
 - C) Hardware configuration
 - D) Programming language

Answer Key (MCQs)

1-A, 2-D, 3-B, 4-D, 5-A, 6-D, 7-A, 8-B, 9-A, 10-A,
11-A, 12-B, 13-A, 14-A, 15-A, 16-A, 17-A, 18-B, 19-B, 20-B